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SPECIAL STRATEGIC PARTNERSHIP ANNOUNCED

Trade, corridor, defence: Rome, Delhi upgrade ties

Modi, Meloni discuss ‘complex scenarios’, chart a new roadmap

Shubhajit Roy
New Delhi, May 20

ELEVATING THEIR bilateral ties to a “special strategic partnership”, India and Italy Wednesday set a trade target of 20 billion euros by 2029, agreed on a defence industrial roadmap, and concluded pacts on critical minerals, countering money laundering and terror financing.

These decisions were taken after Prime Minister Narendra Modi held talks with Italian Prime Minister Giorgia Meloni

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Prime Minister Narendra Modi and his Italian counterpart Giorgia Meloni in Rome, Wednesday. ANI; RELATED REPORT, [PAGE 5](#)

Not in favour of clearing new hydel projects in Upper Ganga region: Centre to SC

Nikhil Ghanekar
New Delhi, May 20

CITING THE “incomparable position” of the Alaknanda-Bhagirathi river basins due to hydrological, ecological and cultural significance, the Centre on Wednesday submitted to the Su-

preme Court that it is “not in favour” of permitting any new hydroelectric projects in the upper reaches of the Ganga river.

It said only the seven existing projects — Tehri Stage-II (1,000 MW), Tapovan Vishnugad (520 MW), Vishnugad Pipalkoti

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Govt pitch to US nuclear mission: Scale up capacity, small modular reactors key

Anil Sasi
New Delhi, May 20

ALONGSIDE A stepped-up role for the private sector in nuclear plant operations, India is doubling down on its domestic civil nuclear programme based on its mainstay, pressurised heavy water reactors (PHWRs), and remains open to collaborating with external players to scale up projects built on this technology.

Less than six months after the landmark legislation opened up India’s nuclear sector, a visiting high-powered American nuclear delegation has been informed, in meetings with top government functionaries, of New Delhi’s two clear

E. EXPLAINED

America’s mission

The US nuclear ‘executive mission’ has a twin-pronged brief: take stock of India’s nuclear energy landscape after a landmark legislation opened up this sector; communicate the US industry’s interest, as well as coordinate US government messaging, on emerging opportunities.

objectives following the legal changes: scale up nuclear power to step up base-load capacity, and progressively enter

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INDIAN EXPRESS UPSC IAS EDITION HD 21~05~2026

-:FOR UPSC IAS ASPIRANTS:-

"AVOID POLITICAL & IRRELEVANT ARTICLES"

Please Try To Read This Completely in 40 Minutes If You Can't So Then You Have To Increase Your Efforts

All the topics of this UPSC IAS Edition are directly or indirectly important for the prelims & main examination. There are some topics which can be coded in answer writing of other topics in the main exam.

Rubio to arrive on May 23 for 4-day visit; trade, energy, defence on agenda

● CIVIL SOCIETY GROUPS OPPOSED TO PROPOSED ‘PRIVATISATION’ OF POWER SECTOR IN THE UNION TERRITORY

May 22 Ladakh talks: Why cracks in key Leh body give Centre an upper hand

Bashaarat Masood
Srinagar, May 20

DAYS AHEAD of the talks between the Centre and Ladakh’s civil society organisations, the rifts in the Apex Body, Leh (ABL) have widened in the wake of its decision to accept the resignation of its founding chairman Thupstan Chhewang, which had been pending for months.

Amid growing differences within the ABL, Chhewang had resigned as its chairman in July 2025, citing health concerns. His resignation was earlier neither accepted nor rejected by the ABL leadership, which finally decided to accept it early this month. The ABL and the Kargil Democratic Alliance (KDA) are the umbrella organisations of various civil so-

ciety bodies in the Leh and Kargil regions, respectively. They are also part of the High Powered Committee (HPC), headed by Union MoS for Home Nityanand Rai, which has convened a meeting on May 22. The HPC has been examining demands raised by the Ladakh representatives.

While the ABL leadership has accused Chhewang of toeing the BJP line, the cracks in the organisation is likely to give the BJP-led Centre the upper hand in the talks. While Chhewang and ABL co-chairman Chering Lakruk had long been at loggerheads, the latest flashpoint between them is the civil society bodies’ opposition to the proposed “privatisation” of the Ladakh Power Development Department (LPDD).

Chhewang has castigated the ABL’s stance on the issue, claiming that it fell outside the latter’s mandate. “He (Chhewang) was saying LPDD isn’t our issue and we shouldn’t raise it, but all others thought it’s important to safeguard the interests of Ladakh,” Lakruk said. “Since his stand seems pro-government, all the members said that we should accept his resignation.”

Both the ABL and KDA have been up in arms against any proposal by the Ladakh administration to “privatise” the power sector in the region. They have held that if the proposal to form a joint venture between the LPDD (49% share) and the Rural Electrification Corporation (51% share), is implemented, it will “worsen” the unemployment problem in the UT, lead to increase in power tariff, and “compromise national



The High Powered Committee, headed by Union MoS for Home Nityanand Rai, has convened a meeting on Friday

and strategic security interests”. In a letter to L-G Vinai Kumar Saxena, they have called for scrapping of the proposal.

A former BJP MP from Ladakh, Chhewang has questioned his removal from the ABL, asking how he could be

ousted when he was the founding chairman of the body. He has also questioned the ABL’s functioning, saying that the talks between the Centre and the Ladakh representatives have hit a dead end. “Both sides have remained firm in their respective

positions and are unwilling to compromise resulting in complete deadlock in the dialogue process,” he told a press conference in Leh a few days ago. “The people are left with two options — either to intensify the agitation for their demands or enter negotiations and accept the proposals the government may eventually be willing to offer.”

Chhewang said the ABL should restrict itself to speaking on the issues which were agreed upon. “We had agreed to speak only on the four-point agenda related to Ladakh’s constitutional and political demands,” he said, adding the issue of privatisation of electricity in Ladakh was not part of the organisation’s mandate. “After I expressed my concerns about it through a message, I was removed. I am the founding

member of the Apex Body — how can I be removed without any proper consultation.”

Sources said Chhewang is upset about the growing clout of the KDA and the representation given to Muslims in the ABL. Lakruk, however, says the agenda for talks was set by Chhewang. “He set the agenda himself. How can he today say that it is a hard stand,” Lakruk asked. “The (joint) platform of ABL and KDA was also formed by him.” The infighting within the ABL is likely to cast a shadow on the May 22 meeting.

The civil society groups from Leh and Kargil have jointly demanded full statehood for Ladakh and its inclusion under the Sixth Schedule, arguing that the UT’s over 90% Scheduled Tribe (ST) population requires constitutional safeguards over land, re-

sources and cultural identity. In September 2025, four persons were killed and many injured in police firing in Leh as protests over demands for statehood and Sixth Schedule protections turned violent. While Chhewang may be alone in his fight against the ABL leadership, a new outfit, Voice of Buddhist Ladakh (VBL), formed early this year amid allegation of it being a “BJP proxy”, could throw its weight behind him. The VBL has claimed to represent Buddhist interests in negotiations with the Centre.

The ABL has alleged that the VBL had been propped up by the administration to undercut the joint movement of the Leh and Kargil organisations. These divisions within the Ladakhi leadership may weaken the bargaining power of its representatives in their dialogue with the Centre.

Boost to India-South Korea defence cooperation as Rajnath visits Seoul

Amrita Nayak Dutta
New Delhi, May 20

DEFENCE MINISTER Rajnath Singh on Wednesday met his South Korean counterpart Ahn Gyu Back in Seoul, where both sides reviewed the entire spectrum of defence cooperation and discussed ways to further expand collaboration in areas such as industry, production, maritime security, emerging technologies, military exchanges, logistics and regional security.

Singh also met South Korean Minister for Defense Acquisition Program Administration Lee Yong-chul and both agreed to harness the symbiotic efforts to create avenues for joint development, production and exports.

A roadmap to unlock the po-

tential of the India-Korea Defence Innovation Accelerator Ecosystem (KIND-X) to synergise the innovation ecosystems of the two countries was discussed.

In a statement, the Ministry of Defence said both sides acknowledged the growing convergence between India's Act East Policy and the Republic of Korea's (RoK) regional strategic vision, "reiterating their commitment to strengthening defence ties in line with the shared objectives of maintaining a free, open, inclusive and rule-based Indo-Pacific".

The ministry said MoUs were inked in key areas of defence co-operation, reflecting the expanding scope and depth of the bilateral partnership.

Agreements were exchanged on promoting cooperation in the



Defence Minister Rajnath Singh with his South Korean counterpart Ahn Gyu-back in Seoul on Wednesday. ANI

field of defence cyberspace; training between India's National Defence College and Korea National Defence University; and UN Peacekeeping Cooperation, mak-

ing the partnership stronger and multidimensional.

In a post on X, Singh said the meeting was highly productive. Talking about the various agreements exchanged, he said this makes the India-South Korea partnership stronger and multidimensional.

Singh is currently on an official visit to South Korea and Vietnam. Earlier in the day, he laid a wreath at the National Cemetery of Korea and paid homage to the soldiers who made the supreme sacrifice in the service of their nation.

He later chaired the India-RoK Defence Industry Business RoundTable, which brought together senior government officials and leading defence industry representatives from both countries. "The interaction provided an

important platform for exploring new opportunities in defence manufacturing, co-development, co-production and supply chain partnerships," the Ministry statement noted, adding that Singh invited Korean defence companies to strengthen engagement with Indian industry and contribute towards long-term mutually beneficial collaboration.

He also highlighted that defence manufacturing is no longer confined only to conventional platforms and equipment, as modern defence ecosystems are powered by advanced electronics, artificial intelligence, autonomous systems, cyber technologies, sensors, semiconductors, quantum technologies, advanced materials and space-based capabilities.

CUTTACK’S GI-TAGGED SILVER FILIGREE INDUSTRY IS STARING AT A CRISIS

Soaring prices, import curbs push Odisha’s filigree artisans to the brink

Sujit Bisoyi
& George Mathew
Bhubaneswar, Mumbai,
May 20

FOR THE better part of the last three decades, Ajay Dey’s routine was unvarying. He would complete his morning routine and head for his family’s filigree workshop in Cuttack’s Shaikh Bazar at 10 am, spending hours crafting silver into intricate items — from the famous Konark wheel and dancing peacocks to delicate necklaces, earrings and bracelets. The hours were long, the work tiring, and for all their labour, this family of artisans would make Rs 30,000-40,000 a month.

But the recent government curbs on silver imports threaten to upend a livelihood already shaken by soaring prices of the precious metal. “I have been working here since I was 15, but I’ve never seen such a dull market before. Our business has already been affected due to a steep increase in the silver price. Now the import duty hike would further worsen the situation. We have not been receiving orders while those who paid in advance have requested not to proceed with the order,” the 44-year-old Dey tells *The Indian Express*.

He isn’t alone in this dilemma. Last week, the central government decided to raise the customs duty on imports of gold and silver to 15% from 6% and eventually put silver imports for domestic consumption under the restricted category, meaning prior government approval is required. The development comes at a time when a persist-



For nearly 500 filigree artisans in Cuttack, the latest government measures spell bad news. FILE

ent supply deficit and higher industrial demand, coupled with geopolitical tensions, have sent silver prices soaring.

For the nearly 500 artisans of the Geographical Indication-tagged Tarakasi artwork in Odisha’s Cuttack, the latest measure spells bad news, especially as the festival season from September to March approaches. Markets that would normally be bustling with activity now look sparse.

“We used to process 15-20 kg of silver every month, which came down to 6-7 kg in the last few months. These days, we hardly process 1-2 kg of silver for filigree work due to lack of demand,” Dey says.

A craft under strain

A 1,000-year-old city, Cuttack is famous for its Rupa Tarakasi (silver filigree) work of intricate design and fine craftsmanship. In Odia, “tara” means wire and “kasi” means to design.

As part of Rupa Tarakasi, silver bricks are transformed into thin fine wires or foils and used to create jewellery and artefacts. The famed handicraft work received a geographical indication (GI) tag in March 2024.

Though the exact origin of the filigree art in Cuttack is unclear, it is known to have existed as far back as the 12th century, with the art form receiving considerable patronage under the Mughals. Typically, Tarakasi is an art form passed down through generations, artisans learning it from family elders and supplying products to jewellery shops across the country and to the state-owned agency — Utkalika. The grand silver filigree tableaux are also popular in puja pandals.

Even before the central government’s measures and soaring silver prices, duplicate filigree products had led to significant attrition in the artisan community. With younger generations

from artisan families opting for other career options, official sources put the number of artisans now at 500 – a quarter of the 3,000 that existed in 1996.

Silver prices have shot up by over 200% since January 2025, with demand boosted by sectors such as solar energy, electric vehicles and semiconductors. The austerity measures sound a death knell for an already shaky industry. According to artisans, silver, which cost Rs 1 lakh per kg last May, now stands at Rs 3 lakh.

Soaring silver prices have also led to a sharp rise in the cost of filigree items. “The price of products has almost doubled in the past one year. Why would someone opt to buy a filigree artefact at a high cost,” said Tabrez Khan, a Cuttack-based artisan, adding that corporate orders have also declined.

Meanwhile, experts insist the restrictions mean India has made silver imports more controlled. “The silver import restriction doesn’t mean India has shut the door, it means the entry is now guarded. Supply isn’t stopping, it’s being channelled only through nominated agencies like the RBI-approved banks, DGFT-approved entities, jewellers via the bullion exchange,” said Jateen Trivedi, VP Research Analyst - Commodity and Currency, LKP Securities. “For the domestic market, that almost always translates into higher premiums,” he said.

His words, however, offer little solace to Dey, who fears for his livelihood. “We can’t do any other work. How would we survive, run our family and repay the loan we have taken?”

Europe’s shift to centre of Delhi’s frame is welcome

PRIME MINISTER Narendra Modi’s trip to the Netherlands, Sweden, Norway and Italy underscores how far India’s perception of, and engagement with, Europe has evolved. Throughout the Cold War, India’s approach to Europe was shaped largely by its close economic and defence ties with the Soviet Union and its rivalry with the West. A prolonged neglect is now giving way to expanding cooperation. Buoyed by the India-EUFTA signed in January this year, following the 2024 India-EFTA agreement, Europe has been increasingly entering India’s strategic imagination despite the friction around Russia’s invasion of Ukraine. The shift comes at a time when traditional alignments are under growing strain — an unpredictable American president, an ever-deepening partnership between India’s ally Russia and competitor China, and a fragile détente between Beijing and Washington, reflected in the Trump-Xi summit.

Europe is now central to India’s pursuit of export markets, capital, advanced technology and green-energy partnerships. The PM’s tour itinerary reflects the enormous potential that collaboration with some of the continent’s smaller but highly capable economies can unlock. The Netherlands, with a population of just 18 million, is a pivotal player in the global chip race because of ASML, the sole manufacturer of the advanced lithography machines needed to produce the most advanced semiconductors. The deal between ASML and Tata Electronics will support scaling up of the latter’s fabrication plant in Dholera. In defence, India seeks to diversify procurement away from Russia, and in renewable energy, it requires both technological expertise and capital. Europe also offers opportunities for India’s youth as the diaspora across the continent continues to expand, driven by highly skilled workers and students. For Europe, meanwhile, India is a critical partner in its efforts to de-risk from a rising China.

While the EU functions as a bloc of 27 member states, each country retains its own strategic priorities and foreign policy outlook. Europe’s sub-regions, too, differ sharply in their perceptions of external powers. India already enjoys strong partnerships with key western European powers such as France, the Netherlands and Germany, and it has moved to deepen engagement with the Nordic countries while strengthening ties with Italy in the Indo-Mediterranean. More must be done to engage with these sub-regions and cultivate cross-linkages for mutual gains. With geopolitical realignments, a fraying rules-based order, and the decreasing relevance of international institutions, India’s ties with Europe can no longer be peripheral to its foreign policy.

In Manipur, pull back from edge, build trust

AFTER MONTHS of brittle calm, Manipur could once again be on the edge. The immediate provocation — the ambush and killing of three Kuki church leaders travelling back to the Kuki-Zo majority Kangpokpi district — has degenerated into a broader atmosphere of fear. There have been retaliatory abductions, disappearances and mounting tensions between Kuki and Tangkhul Naga communities across the hill districts. In a state mired in ethnic conflict for the last three years, and burdened by layers of historical grievance, incidents such as these could turn into communal flashpoints.

The present crisis, that began in Ukhrul earlier this year, however, cannot be understood through the lens of the violence between the state’s Meiteis and Kukis that has riven Manipur since May 2023. It can be traced back, instead, to the older, more serrated Naga-Kuki fault line, shaped by decades of territorial contestation, insurgent rivalry and competing visions of political belonging. The Nagas remain among the Northeast’s most politically influential communities, with powerful tribal bodies, longstanding armed networks, and aspirations tied not merely to demands of autonomy within Manipur, but to a wider Naga political settlement that transcends state boundaries. Instability in the hill districts, therefore, echoes far beyond Manipur, intersecting with unresolved questions of autonomy, territory and ethnic representation. The Kukis, meanwhile, carry their own histories of displacement and grievance. In an already militarised landscape, where weapons continue to circulate freely and trust in state authority has eroded sharply, these unresolved antagonisms threaten to pull Manipur back into another cycle of violence.

The return of an elected government led by Yumnam Khemchand Singh in February had raised cautious hopes of rapprochement in a state exhausted by prolonged conflict. Presented as a conciliatory figure capable of balancing Meitei, Kuki and Naga interests, Singh cannot afford to fall back on the political equivocations of his predecessor. His inclusion of both Kuki and Naga deputy chief ministers offers an opening, but representation alone cannot substitute for trust. His government must move beyond reactive measures towards sustained political mediation: Securing the unconditional release of hostages, guaranteeing safe movement across districts, prosecuting the guilty and establishing credible inter-community dialogue mechanisms. This moment must be decisively defused, too much is at stake.

Our highway policy needs a green reset



VINOD B MATHUR, RAJKUMAR SAPRA AND VINOD KUMAR

INDIA IS expanding its highways at an unprecedented pace. Under Bharatmala and Vision 2047, thousands of kilometres of expressways and highways are reducing logistics costs and accelerating economic growth. This transformation is creating an ecological and public-health burden. Highway corridors are increasingly becoming zones of pollution, landscape fragmentation and agricultural risk. Since many highways pass through densely populated agricultural regions, the impacts extend beyond the carriageway itself.

Highways must no longer be viewed merely as transport infrastructure, but as ecological and socio-economic landscapes. One promising solution is Highway Farm Forestry (HFF) — a market-linked agroforestry approach in which farmers adjoining highways voluntarily cultivate commercially viable tree crops instead of pollution-sensitive food crops. These “Tree Crop Buffer Zones” can reduce pollution exposure, improve ecological connectivity and generate additional income. Unlike earlier social-forestry programmes, HFF is profitability-driven. Species such as poplar, bamboo, teak, Malabar neem and gamhar have strong industrial demand. With assured buyback and extension support, roadside agroforestry can become a viable rural enterprise.

India imports nearly Rs 70,000 crore worth of wood and wood products annually. HFF could reduce import dependence, create rural employment and strengthen timber industries in line with the “Make in India” agenda. However, the proposal is not simply about timber production. It is about integrating ecology planning into infrastructure policy. Conventional roadside green belts are difficult to scale across the highway network, given the high cost of acquiring land. A more practical approach is a differentiated “Tiered Zone Framework”. Highways close to pollution-affected belts — roughly within 100 m — should prioritise tree-based systems, while areas beyond can continue mixed agroforestry or conventional agriculture depending on ecological and livelihood conditions. This approach recognises that ecological restoration, food security and farmer livelihoods must go together.

Properly designed roadside tree systems can function as ecological stepping stones in fragmented landscapes, supporting pollinators, birds and local microclimates. Plantation biomass can also support carbon sequestration and help farmers participate in emerging carbon markets. At the same time, ecological safeguards are essential. Monocultural plantations could create new environmental problems if poorly designed. Species selection must be science-based, region-specific and groundwater-sensitive. Invasive species should be excluded and native and mixed-species systems should be encouraged. HFF is not a substitute for cleaner mobility systems or stronger environmental safeguards in infrastructure planning; it must complement them.

Implementation will require institutional coordination and farmer support. Farmers need quality planting material, extension services, predictable markets and simplified regulations. The National Transit Pass System introduced in 2023 has already addressed a major bottleneck by enabling seamless interstate movement of timber through a “One Nation, One Pass” framework. India now needs a national policy vision that treats highway corridors not merely as engines of mobility, but as landscapes for ecological and economic resilience.

Mathur is former chairperson, National Biodiversity Authority of India. Sapra is former managing director, Haryana Forest Development Corporation, Kumar is former principal chief conservator of forests, Haryana

Highways must no longer be viewed merely as transport infrastructure, but as ecological and socio-economic landscapes. One promising solution is Highway Farm Forestry

The Ideas Page

THURSDAY, MAY 21, 2026

US vs Iran shows why it is not about military power, but manipulating risk



THAROORTHINK
BY SHASHI THAROOR

AS THE “phony peace” in Iran drags on, I am reminded of an old lesson: In the annals of strategic thought, few insights have proved as enduring as Thomas Schelling’s in his 1960 book *The Strategy of Conflict*: Coercive bargaining is not about the blunt application of force, but about the manipulation of shared risk. In conflicts where neither side can afford outright defeat, Schelling showed that the real contest lies in shaping the environment of danger — raising the costs of escalation, narrowing the exits, and forcing adversaries to calculate how much risk they can bear before they must compromise.

The Trump administration’s approach to Iran seemed to rest on the assumption that sufficiently severe bombardment by its overwhelmingly superior military force would compel capitulation. Yet history suggested otherwise. Severe punishment, when it fails to break an opponent’s will, does not produce submission; it produces a bargaining environment in which both sides become desperate to find a way out that does not humiliate them fatally. In such an environment, the weaker party has every incentive to make the exit as costly and as visible as possible, ensuring that the stronger adversary pays a reputational and strategic price for its miscalculation.

Iran’s position in this confrontation reflects Schelling’s logic. Strategically weaker than the US in conventional terms, it nonetheless possesses asymmetric leverage. The Strait of Hormuz is not merely a shipping lane; it is, in Schelling’s terms, a hostage whose value rises as American desperation increases. By causing disruption of global energy flows, Iran can manipulate the shared risk that binds both sides, forcing Washington to reckon with the costs of escalation not just in military terms but in

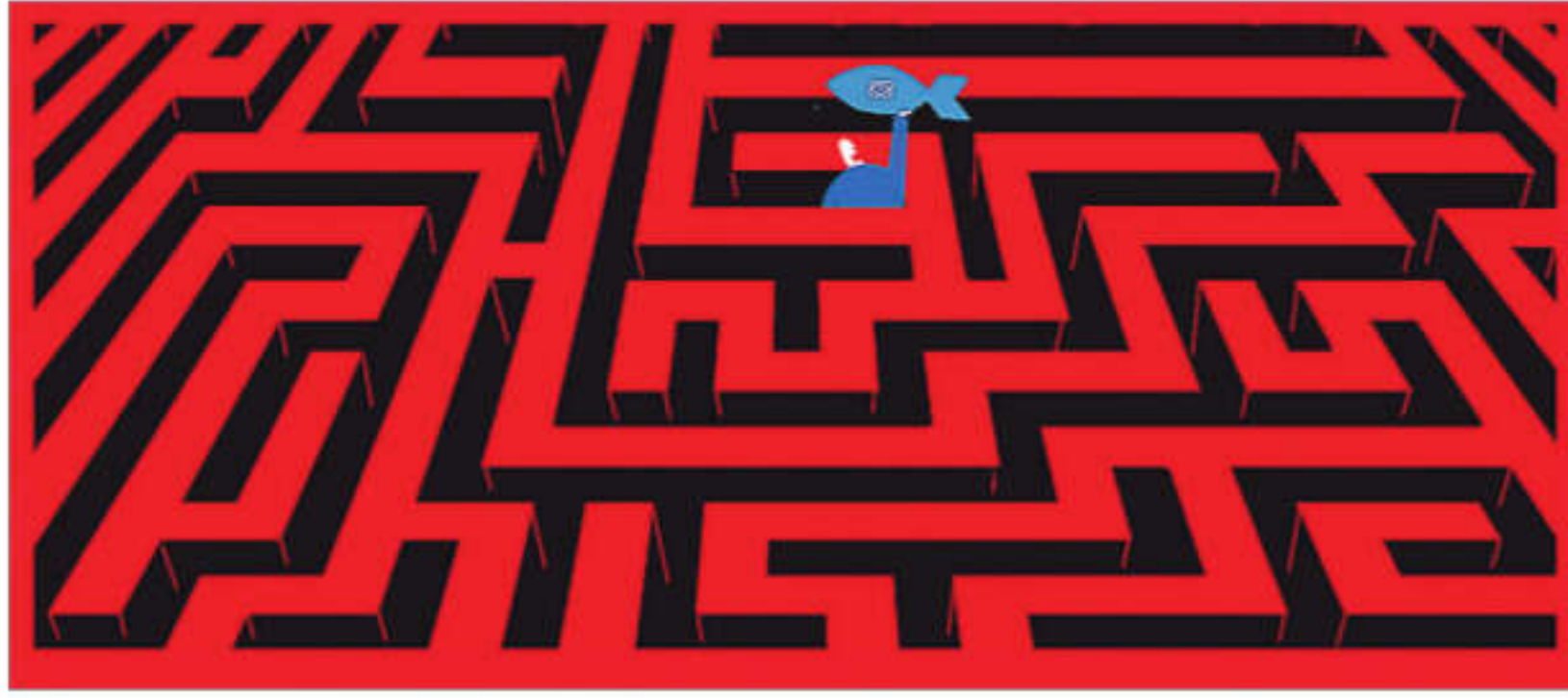


ILLUSTRATION: C R SASIKUMAR

economic and diplomatic fallout. The more the US insists on punishment without offering a credible off-ramp, the more valuable that hostage becomes.

This dynamic underscores a broader truth about coercive diplomacy. Force alone rarely secures compliance. What matters is the structure of risk, the perception of danger, and the credibility of exit strategies. When punishment is pursued without a parallel effort to manage risk and provide pathways to de-escalation, it hardens resistance rather than softening it. Iran’s tactics — calibrated disruption, visible defiance, and the threatened exploitation of chokepoints — are designed precisely to exploit this gap, turning vulnerability into leverage.

History offers ample illustrations. The Cuban Missile Crisis of 1962 was not resolved by America’s overwhelming military superiority alone. It was resolved by the manipulation of shared risk: Both Washington and Moscow understood that escalation could lead to nuclear war, a catastrophe neither could risk. The crisis ended when both sides found a way out that preserved dignity — removing missiles from Cuba in exchange for a quiet withdrawal of American missiles from Turkey. The lesson was clear: Coercion succeeds not when one side overwhelms the other, but when both sides per-

By causing disruption of global energy flows, Iran can manipulate the shared risk that binds both sides, forcing Washington to reckon with the costs of escalation not just in military terms but in economic and diplomatic fallout

ceive that the risks of continued confrontation outweigh the costs of compromise.

Closer to home, the Kargil conflict of 1999 demonstrated a similar dynamic. Pakistan’s incursion into Indian territory was militarily unsustainable, but Islamabad sought to leverage asymmetric risk by internationalising the conflict and threatening escalation. India’s calibrated response — limited military action combined with diplomatic pressure — was designed to manage risk while preserving credibility. The eventual withdrawal of Pakistani forces was not simply a product of battlefield losses, but of the recognition that the risks of escalation, including the nuclear option, were intolerable. That shaped the outcome: India won.

The Gulf War of 1990-91 also illustrates the paradox of coercion. Saddam Hussein’s invasion of Kuwait was met with overwhelming American military power, but the coalition’s success lay in using force to create a bargaining environment where Iraq’s risks mounted daily. The destruction of infrastructure, the isolation of Iraq diplomatically, and the threat of further escalation created a situation in which Saddam’s position became untenable. Yet even then, that war ended with Saddam still in power, a reminder that coercion rarely produces total capitulation. The absence of a credible exit strategy left

the conflict unresolved, sowing the seeds for future confrontation a decade later.

These examples highlight the enduring relevance of Schelling’s insight. Coercion is not a mechanical process of applying force until the adversary breaks. It is a psychological and political process of shaping perceptions of risk, creating incentives for compromise, and managing the pathways to de-escalation. When punishment is pursued without attention to these dynamics, it produces resistance, escalation, and unintended consequences.

The Strait of Hormuz thus stands as a symbol of the paradox of coercion. It is a narrow waterway, but in strategic terms it is vast — a “hostage” whose fate shapes the calculations of nations far beyond the immediate conflict. It is not a mere shipping lane, but the embodiment of shared risk, a reminder that in the strategy of conflict, the manipulation of danger is often more decisive than the application of force.

For policymakers, the lesson is sobering. Military superiority does not guarantee compliance. Punishment without pathways to compromise creates bargaining environments where adversaries exploit risk to their advantage. The challenge is to integrate force into a strategy that recognises the psychology of risk, the importance of dignity, and the necessity of credible exits. Without this, coercion becomes a trap, locking adversaries into cycles of escalation that neither can afford to sustain.

In this case, both Iran and the United States must recognise that the manipulation of risk is the essence of the contest. Severe bombing may demonstrate power and resolve, but without a credible off-ramp, it only increases the value of the other side’s capacity to cripple the world economy. The Strait of Hormuz will remain a lever of asymmetric power, and Iran will continue to exploit it until Washington offers a pathway to compromise that preserves dignity on both sides. Schelling’s insight, six decades old, remains the key to understanding why mere coercion fails, and why shared risk is the true currency of bargaining in international conflict.

The writer is a fourth-term Member of Parliament (Lok Sabha) for Thiruvananthapuram and chairman of the Parliamentary Standing Committee on External Affairs



AMRITA NANDY

IN THE award-winning Lebanese film, *Capernaum*, the 12-year-old protagonist, Zain, is asked by the judge why he wants to sue his parents. “Because I was born,” Zain says. Through his parents, Zain confronts us, adults, for first turning the world into Capernaum (broadly, a site of chaos) and then bringing children into it.

In the recent debate about the right to abortion beyond 24 weeks, Zain could be any one of us, the unborn child of the 15-year-old rape survivor, or its 15-year-old child-mother. In a remarkable departure from the law and the opinion of the medical establishment, the Supreme Court (SC) permitted abortion.

Among the opinions sparked by this permission, a recent article (‘Law is not the problem. Abortion in India needs honest engagement and implementation’, *IE*, May 15) mentioned the “spiritual dimension” of the potential birth of the unborn child. Anti-abortion discourse often leans on the religious-spiritual to support its arguments for birth.

However, there are different spiritual dimensions at play as well — for the right of the child not to be born. “Spiritual” here carries its humanistic, existential connotations that centre our shared fibre and invoke our innate oneness. Like Zain, millions are born — many accidentally — in circumstances that are unfit for a child, if not precarious or harmful. It is this unnecessary yet pervasive human suffering that Zain chal-

The grip of pronatalism and motherhood on our psyches can blind us to her sense of being overwhelmed around selfhood, existence, and purpose

IN GOOD
faith

lenges in the fictional court. By allowing the termination of the pregnancy of the minor, the SC has prevented many real-life Zains from reaching their overcrowded courtrooms. The point? Birth is not necessarily a welcome event, even via spiritual and philosophical frames.

This is an upstream argument, and therefore even more fundamental to the debate on abortion. Since suffering is an inescapable reality of human life, every newborn will inevitably suffer repeatedly throughout their life. Suffering implies a range of psychological distress caused by dissatisfaction, anxiety, anger, hatred, depression, besides bodily pain and discomfort. Philosophers such as David Benatar (author of *Better Never to Have Been*) see birth as a moral wrong because it places an innocent being in harm’s way without their knowledge and consent. But since the parent knows that the newborn will suffer, they are accountable for their choice to give birth. This act of harm is driven by a spectrum of the parents’ own present and future needs of caregiving and receiving.

This cynical perspective is not just held by a global, growing community of antinatalists, the centrality and certainty of suffering are also affirmed by spiritual traditions such as Buddhism. It calls birth the first step in recurrent cycles of suffering that include sickness, ageing, and death. Birth unleashes a continuum of *trishna* (craving), which creates a continuum of suffering that keeps

ustrapped in the world. Buddhism is not antinatalist and does not prohibit procreation for its lay followers. But its ultimate recommendation against the suffering of human life — or “crossing the flood” (*Samyutta Nikāya* or Connected Discourses) — is to transcend birth itself. It can be done by cultivating discerning self-awareness and weeding out self-deception, delusion, and craving. This could mean, including for the issue of abortion, a balanced response between compassion for the raped minor and her circumstance and a fixed, doctrinaire refusal to abortion.

An equally potent consideration is the spiritual distress of the pregnant person, in this case, a child herself. The grip of pronatalism and motherhood on our psyches can blind us to her sense of being overwhelmed around selfhood, existence, and purpose. Bringing a new human into the world, under any circumstance — even through the mother’s strong unwillingness and anguish, or coercing her to carry the child to term only to offer the baby for adoption — is a fraught proposition that does not augur well for either the parent or the potential child. There is ample evidence about the relationship between unwanted pregnancies, childhood trauma, abuse, and lifelong psychological challenges for the mother and child.

Between an unborn child and its living mother, the Court chose well not to reproduce suffering from one to two.

Nandy is author of Motherhood and Choice: Uncommon Mothers, Childfree Women

Xi, Putin unite to criticise US, but fail to clinch big gas pact

Jointly condemn Trump's Golden Dome shield plan

Reuters
Beijing, May 20

CHINA AND Russia condemned US President Donald Trump's Golden Dome missile defence shield plans and Washington's "irresponsible" nuclear policy at a summit on Wednesday, a week after President Xi Jinping hosted Trump in Beijing.

A joint statement issued after Xi's summit with Russian President Vladimir Putin served to underline that, while the Chinese leader seeks stable and constructive relations with Trump, he differs fundamentally with him on key issues where China's position is closely aligned with Russia's.

The statement said Trump's plan for a ground- and space-based missile interceptor system threatened global strategic stability and criticised Washington for allowing a treaty restricting the US and Russian nuclear arsenals to expire.

The treaty lapsed in February and Trump did not respond to Moscow's proposal to extend its missile and warhead limits by a year, something that some US politicians argued would have hindered the US from responding to a nuclear build up by China.

Yet while speaking in uni-



Russian President Vladimir Putin and Chinese President Xi Jinping visit an exhibition by the TASS and Xinhua news agencies at the Great Hall of the People, in Beijing on Wednesday. AP

son on global security issues, the two leaders failed to reach a breakthrough that Moscow has long been seeking - a contract for a new pipeline that would enable it to more than double the amount of natural gas it sells to China. Xi was wrapping up a remarkable week of diplomacy in which he met the leaders of China's most powerful strategic rival and one of its closest partners.

With Trump seeking an exit from a war with Iran and Putin's forces largely bogged down in Ukraine, the summits provided China's leader with a chance to showcase Beijing as a pillar of global stability and an indispensable diplomatic player.

"Xi does appear to hold the stronger position relative to

both Putin and Trump.

Both leaders are grappling with conflicts of their own making that have proven far more difficult to resolve than initially anticipated," said Patricia Kim, a foreign policy fellow at the Washington-based Brookings Institution.

"Xi, meanwhile, has been able to focus more squarely on strengthening China internally while projecting an image of a stable and confident great power on the global stage."

While the summit with Trump was largely about managing tensions, the encounter with Putin posed a different challenge - how to demonstrate progress in a relationship that the two sides have already proclaimed is "without limits".

DESPITE BEING MOST POPULOUS STATE, UP's GOLD LOAN OUTSTANDING WAS JUST RS 42,300 CR

Five southern states account for 75% of outstanding gold loans

Gold loans have emerged as the fastest-growing segment in the lending market, surging 50.4% on-year

George Mathew
Mumbai, May 20

FIVE SOUTHERN states account for nearly 75% of the country's total gold loan outstanding, underscoring the region's dominant role in a market that has witnessed explosive growth over the past two years, says a report.

Out of the total Rs 18.6 lakh crore gold loan outstanding (including banks and non-banking finance companies), five southern states — Tamil Nadu, Andhra Pradesh, Karnataka, Telangana and Kerala — account for Rs 13.94 lakh crore as of March 2026, according to data prepared by CRIF High Mark, a credit information company.

Tamil Nadu leads with Rs 5.96 lakh crore outstanding, Andhra Pradesh follows with Rs 3.08 lakh crore, Karnataka with Rs 1.81 lakh crore, Telangana at Rs 1.60 lakh crore and Kerala Rs 1.45 lakh crore as of March 2026, it said.

Driven by rising gold prices, increasing credit demand, and the growing preference for secured borrowing, the gold loan segment has expanded rapidly, with southern India continuing to be the industry's stronghold due to its deep cultural affinity with gold ownership and an extensive network of lenders operating in the region.

Incidentally, despite being India's most populous state, Uttar Pradesh accounts for gold

loan outstanding of only Rs 42,300 crore. Other large states also report relatively modest figures. West Bengal stands at Rs 35,000 crore, Rajasthan at Rs 41,700 crore and Gujarat at Rs 57,100 crore. These numbers highlight the uneven regional penetration of the gold loan market, with several economically significant and densely populated states still lagging behind southern India in the adoption of gold-backed borrowing.

Growth in March 2026 was led by southern states — Karnataka (10.5%), Telangana (12.8%), followed by UP (11.2%).

"All top 10 states recorded improvement. However, AP and Kerala remained above the pan-India average, especially given AP's higher growth rate of 9.4% — on par with the national average," CRIF High Mark said.

Gold loans sustained robust portfolio expansion, rising 50.4% year-on-year and 15% quarter-on-quarter, supported by higher collateral values and partly driven by the reclassification of agri-gold loans into retail, CRIF High Mark said. The sharp rise in gold prices also prompted people to go for gold loans, helping them to get more finance to meet various expenses.

"It has now emerged as the second-largest product in retail lending after home loans, underscoring its growing relevance in customer credit portfolios. We are also witnessing a

INDIA'S GOLD LOAN MARKET

Tamil Nadu	₹5,96,700 cr
Andhra Pradesh	₹3,08,900 cr
Karnataka	₹1,81,800 cr
Telangana	₹1,60,800 cr
Kerala	₹1,45,800 cr
Maharashtra	₹1,11,200 cr
Gujarat	₹57,100 cr
UP	₹42,300 cr
Rajasthan	₹41,700 cr
West Bengal	₹35,000 cr
Top 10	₹16,81,000 cr
Rest of India	₹1,81,800 cr
Total	₹18,62,800 cr

As of March 2026. Source: CRIF High Mark

shift toward higher ticket sizes, and more income-generating end uses, reflecting evolving borrower needs and increased comfort with loans against gold," said Mohit Jain, Group Head - (Co)Lending, Gold, Micro Loans, Distribution, Financial Inclusion), Axis Bank.

The report said PSU banks continue to dominate originations by value, though their position has softened over time. Their share declined from 51.1% in Q4FY24 to 44.6% in Q4FY26, even as they remain the largest category.

Non-banking finance companies have emerged as the clear growth driver, with originations value rising sharply from 20.7% in Q4 FY24 to 31.6% in Q4 FY26.

In volumes, NBFCs lead with 49% share in originations volume as of Q4 FY26, underscoring NBFC-led market expansion driven by faster growth and



Southern India continues to be the industry's stronghold due to its deep cultural affinity with gold ownership. REUTERS

wider distribution reach. PSUs dominate higher ticket size loans while NBFCs are growing stronger in lower ticket size loans (as indicated in their value vs volume shifts), according to the report.

Gold loans emerged as the fastest-growing segment in the lending market, with portfolio outstanding surging 50.4% year-on-year, reflecting strong consumer demand and increased reliance on secured borrowing. Personal loans also recorded healthy growth of 12.9% on year, underscoring sustained demand for unsecured retail credit despite tighter lending norms, the report said.

The gold loan segment also demonstrated improving asset quality, reinforcing its role as a key engine of retail credit growth in FY26.

"Early-stage delinquency declined across all ticket sizes between March 2025 and March

2026, with larger loans improving the most — the Rs 2.5 lakh-5 lakh bucket fell from 1.42% to 0.76% and Rs 5 lakh plus bucket from 1.43% to 0.73%, reflecting stronger collateral coverage in higher-ticket segments," CRIF High Mark said. Among other retail credit categories, consumer durable loans expanded by 20.8%, driven by rising purchases of electronics and household appliances.

Vehicle financing, including auto and two-wheeler loans, continued to witness steady momentum, registering growth between 13.9% and 15.1% year-on-year amid improving consumer sentiment and mobility needs.

Meanwhile, home loans maintained stable and consistent growth at 9.4%, supported by sustained housing demand and relatively resilient property market activity, according to CRIF High Mark.

War is top political violence risk for firms: Allianz report

George Mathew
Mumbai, May 20

WHAT REPORT SAYS

WAR HAS become the political violence risk that businesses fear the most, overtaking civil unrest, as conflicts in Europe and West Asia disrupt global trade and increase risks for companies' operations and assets, according to annual Allianz Risk Barometer 2026.

In the "gray zone" between peace and open warfare, threat actors are leveraging technology, criminal networks and disinformation to attack critical infrastructure, high-profile companies and individuals, blurring the lines between state and non-state actors while undermining detection efforts, Munich-based Allianz said in its 'Political violence and civil unrest trends 2026' report.

"Deteriorating social cohesion is contributing to rising civil unrest worldwide, with protests triggered by economic, political, and social concerns," it said.

While hardship is a major driver, analysis by Allianz shows unrest is not confined to fragile states, with resilient economies like France and Germany experiencing some of the most frequent protests, it said.

ONGOING CONFLICTS, such as the US-Iran conflict, war in Ukraine and other tensions in the West Asia, have heightened global security concerns, disrupted trade routes, and strained international alliances

THE REPERCUSSIONS of the US-Iran conflict on the political violence and terrorism risk landscape are significant

HEIGHTENED RISK of strikes, riots, and civil unrest is to be expected, particularly in countries heavily reliant on West Asian oil and gas

Although terrorism fell overall in 2025, it surged in the West, driven by polarisation, antisemitism, Islamophobia and far-right activities.

"The war in Iran has intensified the terror risk, while extremist groups in sub-Saharan Africa continue to exploit security vacuums. Meanwhile, algorithmic amplification on digital platforms accelerates youth radicalisation," Allianz report said.

"The world is experiencing

one of the most volatile geopolitical periods in our history since the end of World War Two. These are unsettling times. Even before the current conflict in the West Asia, the risks posed by political violence and civil unrest had been intensifying, evolving into global challenges that impact companies, communities, and the wider economy," said Thomas Lillielund, CEO, Allianz Commercial.

As we navigate this era of heightened uncertainty, understanding the implications of these risks and mitigating them in our interconnected business ecosystems has never been more critical, Lillielund said.

Ongoing conflicts, such as the US-Iran conflict, war in Ukraine and other tensions in the West Asia, have heightened global security concerns, disrupted trade routes, and strained international alliances, the report said.

The rise in unilateral actions, such as the removal of Venezuela's President Nicolás Maduro, could embolden adversarial states, undermining geopolitical and economic stability through 2026 and beyond, it said. **FULL REPORT ON**

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Deepening India-US ties

Union Finance Minister Nirmala Sitharaman met US Ambassador to India Sergio Gor in New Delhi, on Wednesday. They discussed India-US economic and financial partnership, fintech collaboration and bilateral cooperation. X/@FINMININDIA

UN cuts India's 2026 GDP forecast

United Nations: The United Nations has revised downward India's economic growth forecast for 2026 to 6.4% from its earlier projection of 6.6%, citing global uncertainties and economic shocks arising from the ongoing West Asia crisis. As per the report released by the UN Department of Economic and

Social Affairs (UN DESA) on Tuesday, India, however, remains one of the fastest-growing major economies.

West Asia crisis has delivered yet another shock to the global economy, slowing growth, reigniting inflationary pressures and heightening uncertainty, it said. **PTI**

● INFRASTRUCTURE

Behind Kerala govt’s scrapping of SilverLine

Shaju Philip
Thiruvananthapuram, May 20

THE CONGRESS-led United Democratic Front (UDF) Cabinet on Wednesday decided to denotify the semi-high speed rail project, SilverLine, mooted by the previous Left government. Chief Minister V D Satheesan said the government was not opposed to the high-speed rail project, but that the project was “an environmental disaster and financially unviable”.

“UDF had studied the project and found there was not even a detailed project report. The land acquired by the government was not denotified. This has caused problems for the people living in the project area,” he said.

The Rs 64,000-crore project was envisaged to run between Thiruvananthapuram and the northernmost district of Kasargod to reduce the existing travel time from 12 hours to four hours. A high-speed transport system has long been a crucial need, given Kerala’s geography and population density.

The project

The project was slated to be executed by Kerala Rail Development Corporation Ltd. or K-Rail, a joint venture registered by the Kerala state government and Indian Railways in a 51:49 equity partnership. K-Rail was specifically formed to develop railway infrastructure in Kerala. In 2021, the state Cabinet gave sanction and a token allocation of Rs 2,000 crore from the Kerala Infrastructure Investment Fund Board.

The 530 km standard-gauge line was planned to run parallel to the existing railway line and mostly on earthen embankments. The trains were expected to run at a maximum speed of 200 km, and a nine-car rake would carry 675 passengers in two classes. The stretch would have 11 stations en route.

● THE ALTERNATIVE

- In January 2026, the LDF government greenlighted a Regional Rapid Transit System as a practical and suitable alternative to SilverLine.
- The LDF government said an MoU would be signed with the Centre once it gave its in-principle approval.

Opposition to the project

For the project, 1,383 hectares (ha) would have to be acquired for the railway line, including 1,198 ha from private parties. Widespread protests erupted across the state, supported by the Congress and the BJP, with critics raising concerns about the large-scale displacement of families and potential environmental damage resulting from the project.

Activists claimed that the proposed alignment would cut through wetlands, paddy fields, and ecologically sensitive zones, while opposition leaders warned of potential financial losses. The Railways concurred with the latter view, questioning its viability in Kerala, where the six-lane national highway development is under way. It ultimately did not give its sanction for the project, saying the project would affect the expansion plans of the existing railway tracks.

Satheesan justified the denotification, saying the project was abandoned and no transactions could be carried out regarding the lands notified for acquisition. The state had abandoned the project, while the Centre did not give its approval, he added.

● POWER

How thorium as nuclear fuel can help India meet its long-term energy needs



EXPERT EXPLAINS
ANIL KAKODKAR
FORMER CHAIRMAN OF THE DEPARTMENT OF ATOMIC ENERGY

IT IS heartening to see indigenously developed pressurised heavy water reactors (PHWR) getting into the mainstream of India’s energy push towards a net-zero compliant Viksit Bharat. The 100 GWe (gigawatt electric) nuclear energy mission launched by the government is to be realised by 2047.

The high rate of capacity addition necessary for such deployment would need a credible implementation framework and

standardised mature, commercially competitive technologies. PHWRs are a fit case.

The attendant steep increase in uranium imports is, however, a matter of concern. Given the uranium demand-supply mismatch, expected rise in uranium demand and consequent to global nuclear generation growing three to four times, and complexities of nuclear-related geopolitics, nuclear fuel supply security could be at risk in 10-15 years.

A much-needed shift

So, a quicker shift to thorium — that India has in abundance — remains critical to India’s energy independence. Large reactor capacity that can produce uranium-233 (U-233) from thorium at scale while also producing electricity is a prerequisite for this. Thus, the first criticality of the 500 MWe (megawatt electric) prototype fast breeder reactor (PFBR) that would open the path to growing such a capacity through fuel breeding marks a key milestone for India. Thorium can also enable India’s transition to a significant energy exporter from being a major importer.

New developments beyond PFBR would also be necessary for the growth of thorium irradiation platforms. These include reactor and nuclear fuel cycle technologies based on mixed oxide, metallic, and thorium fuels.

The start of thorium utilisation at scale is thus three to four decades away. Pursuing this development is nevertheless critical. Fast reactors breed new fuel. They are necessary to expand energy-generating systems that would run as long as thorium is available.

Nuclear generation capacity through uranium-fuelled reactors is soon expected to cross 10 GWe, the target set for the first stage of the three-stage nuclear power programme. The spent fuel from these plants already constitutes the required inventory waiting to feed the second stage as planned. The experience with PFBR so far suggests that in addition to moving closer

to large-scale thorium utilisation, fast reactors can provide economically competitive electricity without dependence on fuel imports. Nuclear generation capacity of a few hundred GWe can be established in this mode.

Timeline for transition

But what about the timeline for transition from uranium to thorium? The intention behind opening international civil-nuclear co-operation was to remove the hurdles in accessing uranium — priority for thorium was to be speeded up. Access to imported uranium is now expected to make the first stage 10 times bigger, that is 100 GWe. PHWRs could constitute around 60 GWe.

Apart from electricity supply, such a substantive PHWR capacity between now and 2047 would also enable a capacity for production of U-233 at scale through thorium irradiation in PHWRs. This would happen much earlier than build-up of similar capacity with fast reactors. One can see at least a three-dec

ade time difference between the two.

Use of thorium-HALEU (high-assay low-enriched uranium) fuel in place of natural uranium in PHWRs would enable thorium irradiation at scale while leading to several advantages in terms of safety, economy, waste minimisation, proliferation resistance, and some saving in the mined uranium. One can thus start building inventory of spent fuel to produce fuel for third-stage thorium reactors.

In my view, the thorium molten salt reactor (TMSR) operating in the near thermal neutron spectrum offers an ideal choice for the third stage. This would enable significant electricity generation capacity sustainably fuelled by thorium. Rough estimates suggest that PHWRs running on thorium-HALEU fuel could support new TMSR capacity comparable with the mother PHWR annually.

The availability of thorium-HALEU fuel recycle technology for feeding TMSRs and development of TMSRs would need to be ensured in about 10-15 years to start producing

electricity from thorium. These technologies are anyway a part of the technology development agenda for the three-stage nuclear power programme. Advancing their deployment will enable us to get on with thorium utilisation at scale without having to wait for completion of second-stage development and large-scale thorium irradiation therein. With fast reactor capacity scaling up to the required level, one can assure U-233 supply and required energy availability at least till fusion energy arrives on the scene.

The world has restricted the use of uranium only in once-through mode. This is due to the fear of potential for malevolent diversion. utilisable energy potential in uranium thus remains two orders of magnitude lower for such countries. Proliferation resistance of the thorium-U-233 fuel cycle should enable realisation of fuller energy potential of thorium. Thorium systems are thus destined to play a greater role in coping up with base load energy demand of the world. This is a unique opportunity for India.

● CLIMATE

Rising night-time heat an urgent health hazard



AMITABH SINHA

WHILE MOST conversation about extreme heat during Indian summers centres on peak daytime temperatures and heatwave spells, it is slowly becoming evident that warmer nights could pose a greater health risk to human beings, particularly those living in low- and middle-income housing units.

Both day and nighttime average temperatures are rising in India, but recent data suggest that nighttime temperatures may be increasing at a faster rate. Experts say indoor thermal exposure could be a bigger threat to public health than exposure to daytime heat outdoors.

A cooler evening and night allow the human body to recover from the heat experienced during the daytime. But if nighttime temperatures also remain high, the body is unable to recoup, and exposure to heat becomes prolonged and sustained, without any relief. People are confined in small spaces for several hours at a stretch. The situation gets aggravated inside houses without natural ventilation or access to air-conditioning.

A recent study by Climate Trends, a Delhi-based climate-focused research organisation, in 50 houses in Chennai—all medium- and low-income residential units—showed that occupants were frequently sleeping in temperatures exceeding 32° Celsius. Sometimes, night-time temperatures even exceeded 35° Celsius, very similar to peak day-time temperatures in the city. “This study highlights the need for passive cooling solutions and structural changes in low-cost housing in urban areas. In the next phase, we plan to study the damage that indoor heat exposure is doing to our bodies,” said Palak Balyan of Climate Trends, one of the authors of this study.

Rising night-time temperatures

India’s average temperature has increased by about 0.7° Celsius between 1901 and 2018, according to the 2020 comprehensive climate change assessment over the Indian region. Both day and night-time tem



People standing near a shop selling air coolers in New Delhi. ANI

peratures have increased during this time, but at different rates.

In the 30 years between 1986 and 2015, the temperature of the warmest day of the year increased by about 0.63° Celsius, while that of the coldest night increased by 0.4° Celsius. This suggests that the rise in daytime temperature was more pronounced than the nighttime temperatures, which is true for this period.

But the situation changes in future projections. The same assessment said that the temperature of the warmest day of the year could rise by 4.7° Celsius by the end of this century, while that of the coldest night could jump by as much as 5.5° Celsius. Night-time temperatures are all set to rise at a faster rate. This has already begun to happen, according to more recent data by the India Meteorological Department (IMD).

While the rapid rise in night-time temperatures is happening everywhere, it is more pronounced in urban spaces, mainly because of what is called the urban heat island effect. Concrete, roads, bricks and metal absorb the heat during the day and

Relief measures

The Climate Trends study noted there are measures that can bring immediate relief to people living in cramped households.

- Passive cooling solutions like reflective roof coatings or whitewashed roofs and walls can make a lot of difference.

radiate it at night. The lack of adequate vegetation and green spaces, loss of water bodies and dense clustering of high-rise buildings amplify this effect. Increasingly, the use of air-conditioners, which emit hot air outside, is emerging as a significant contributor. There could be a difference of 4-6° Celsius or more in the nighttime temperatures of a city centre or a dense residential area of a city, and its outskirts. As we build more, this effect is likely to get further aggravated.

Most of urban India goes to sleep in small houses that are poorly designed and lack natural ventilation. If these houses lack air-conditioners, there is little nighttime relief for the human body. Sleep discomfort, restlessness, exhaustion, and lack of energy are direct results of this condition, which affects people’s productivity the next day. But there are major adverse health impacts as well, whose magnitude is still being assessed.

Heatwave intermittent, heat perennial

Dileep Mavalankar, former director of

the Indian Institute of Public Health in Gandhinagar who has done extensive work on heat and its health impacts in India, gave an insightful example. He said the Ahmedabad Municipal Corporation maintains a daily record of all-cause mortality in the city.

“Ahmedabad records about 100 deaths every day – deaths due to all causes. We have seen that on days when the maximum day-time temperature exceeds 45 degree Celsius, if the maximum night-time temperature is below 28 degree Celsius, there is not much of a change in the daily all-cause mortality. But if the night-time temperature rises to between 28 and 30 degree Celsius, all-cause mortality increases to about 165. If the night-time temperature is more than 30 degree Celsius, mortality goes up to as high as 265. This is an interesting correlation. Even without doing a full-scale analysis of the cause of death, we can see something very significant is happening here. The thresholds would be different for other cities, but there is a very clear indication that higher night-time temperatures are strongly linked to mortality,” Mavalankar said.

At least 23 states and more than 200 cities now have heat action plans that get implemented every summer. A bulk of that is focused on dealing with heatwave-like situations. But a heatwave is a very specific event and is declared when very specific temperature criteria are satisfied. It happens maybe two or three times a month. Extreme heat, on the other hand, is a constant in Indian summer. And so is the trend of rising night-time temperatures.

One component of most heat action plans is about long-term measures needed to reduce impacts. These include coordinated efforts to improve urban infrastructure, come up with better low-cost housing, and increase green spaces. There is a need for paying greater attention to these aspects of the heat action plan.

The Climate Trends study on Chennai households noted there are measures that can be applied to bring immediate relief to people living in cramped households. Passive cooling solutions like reflective roof coatings or whitewashed roofs and walls can make a lot of difference. Some structural changes, like introducing natural ventilation, can also help.